

Highlights

Three Diagnostics for Trustworthy Efficiency and Reliability Claims in Neural-Coordinate Weighted Ensemble Sampling

M.Y. Alzaatreh, H. Alsenawe

- A 17-seed variance decomposition shows an ensemble's seed-to-seed spread is entirely finite-replica noise, not network-initialization variance.
- An unseeded network initializer silently reversed a headline efficiency claim between two identical reruns.
- Two independently tested schedule extensions (finer binning, larger sampling budget) both fail to rescue the learned coordinate once the seeding bug is fixed.
- Charging real wall-clock inference cost, not simulation-sweep cost, reverses an efficiency comparison for a companion coordinate on the same system.
- We propose a three-item audit checklist requiring no new expensive simulation, only reanalysis of data already collected.

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M.Y. Alzaatreh*, H. Alsenawe

Natural Sciences Unit, Fahad Bin Sultan University, Tabuk, Saudi Arabia

Fahad Bin Sultan University, Tabuk, Saudi Arabia

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ABSTRACT

Reports of machine-learning-accelerated Monte Carlo methods commonly compare a learned sampling coordinate against a hand-picked baseline using a single point-estimate figure of merit (FOM), often computed from simulation-sweep cost alone and from a single trained network. We show, using a Weighted Ensemble (WE) rare-event sampler for Ising and Edwards–Anderson spin systems as a fully worked case study, that this practice can produce headline claims that do not survive three inexpensive diagnostic checks. First, a one-way variance decomposition of a 17-network ensemble shows that the entire apparent seed-to-seed spread in a published efficiency ratio is attributable to finite-replica sampling noise, not to genuine network-initialization variance; the between-seed variance component is exactly zero after correction. Second, an unseeded random network initializer silently reversed a training-objective comparison’s headline result from a +14% win to a –17% loss between two runs of an identical configuration; once seeded and swept over 17 initializations, the comparison’s own natural extensions (finer binning, larger sampling budget) both independently fail to help the learned coordinate, and the properly cost-corrected verdict is a significant loss (FOM ratio 0.405, 95% CI [0.268, 0.623]) rather than the inconclusive result obtained from simulation-sweep cost alone. Third, we summarize a companion result in which charging a learned coordinate’s real wall-clock inference cost, rather than its simulation-sweep cost, converts an apparently favorable comparison into a ~3–5× efficiency loss. None of the three findings required new expensive computation to obtain – each reused already-collected simulation output and cost between minutes and a few hours of additional analysis. We package the three checks as a short, concrete audit checklist for reporting learned-coordinate rare-event sampling results, and argue that it should precede, not follow, any point-estimate efficiency or reliability claim.

1. Introduction

Machine-learning-accelerated Monte Carlo methods are increasingly proposed as replacements or supplements for hand-designed importance functions in rare-event sampling, from neural bias potentials [6] to label-free self-consistency training [7] to committor-network approaches for transition-path problems [8–11]. A typical claim in this literature has the shape: “coordinate *A* (learned) outperforms coordinate *B* (hand-picked) by *X*% on figure of merit *Y*,” supported by one or a handful of runs of each method.

This paper does not propose a new sampling method. Instead, using a Weighted Ensemble (WE) rare-event sampler for Ising and Edwards–Anderson spin systems as a fully instrumented case study, we show that three inexpensive diagnostic checks – none requiring new expensive simulation – can change the verdict of exactly this kind of claim, in one case reversing its sign entirely. The case study is a companion project to a physics-focused paper reporting a validated transport-inspired learned coordinate for the same system [13]; the present paper is methodological, focused on comparisons made and abandoned during that project’s development, together with an audit of a currently used component of it.

The three diagnostics are:

- (1) **Decompose seed-to-seed variance before ensembling to reduce it.** An observed spread across independently initialized networks can be mostly or entirely replica-sampling noise rather than genuine initialization variance, in which case ensembling several networks pays real inference cost to cancel a source of variance that is not there (§3).

*Corresponding author

 mzaatreh@fbsu.edu.sa (M.Y. Alzaatreh)

- (2) **Audit random-seed control, and check that natural schedule extensions do not silently rescue a negative result.** An unseeded initializer can make a single comparison’s sign arbitrary; once fixed, a result should be checked against the schedule knobs a reviewer would naturally ask about, with the mechanism of any failure identified, not just its outcome (§4).
- (3) **Charge a learned coordinate’s real wall-clock cost, not its simulation-sweep cost.** A cost accounting restricted to Monte Carlo sweeps is blind to repeated network inference, which can dominate wall-clock time and reverse an efficiency verdict (§5, summarizing a companion result).

Each diagnostic is demonstrated on real data from the same project, using methods (ANOVA-style variance decomposition, paired reruns, bootstrap confidence intervals) that are standard but, in our reading of the recent neural-importance-sampling literature, not yet routinely applied together before a headline claim is reported. §6 distills the three diagnostics into a short checklist and discusses when each is worth the (small) additional analysis cost it requires.

2. Case-study system

The case study is a Weighted Ensemble (WE) sampler [1–3] targeting deep magnetization or energy tails of the two-dimensional Ising ferromagnet and the Edwards–Anderson (EA) spin glass [4, 5] on $L \times L$ lattices, with a learned neural coordinate used to bin walkers during resampling. Two distinct training objectives for the learned coordinate were developed over the course of the project: a *surrogate-label* objective, in which a network is trained to regress a milestone-style progress label collected from pilot trajectories, and a *self-consistency* objective following Kim and Cai [7], in which the network is trained without labels via a harmonic backward-equation residual. The companion physics paper [13] reports validated results for the self-consistency objective, including a disorder-stratified reliability gain and its associated wall-clock cost.

The present paper does not use the self-consistency results. It instead examines two lines of analysis internal to the same project that never appeared in the companion paper: an ensemble-variance investigation conducted on the surrogate-label objective’s 17-seed sweep (§3–§4), and, briefly, the companion paper’s wall-clock cost correction viewed as a third instance of the same general problem (§5). We refer readers to [13] for the full derivation of the WE estimator, the adjoint/committor correspondence motivating the learned coordinate, and the self-consistency training procedure; only the details needed to follow the diagnostics below are repeated here.

For a WE run with replicas R , the figure of merit used throughout is

$$\text{FOM} = \frac{1}{\text{rel.sd}^2 \cdot \text{cost}}, \quad (1)$$

where rel.sd is the relative standard deviation of the estimated target probability $\hat{\pi}$ across replicas and cost is a per-replica computational cost. The central point of this paper is that *which* cost enters this denominator – simulation sweeps only, or total wall-clock time including network inference – is itself a methodological choice that can change the sign of a comparison (§5).

3. Diagnostic 1: decompose variance before ensembling

3.1. Motivation

A common response to noisy learned-coordinate results is to train several independently initialized networks and pool or ensemble their outputs, on the assumption that network-to-network variability is a meaningful source of noise worth averaging over. This assumption is testable, and testing it is cheap relative to the cost of training and running the extra networks.

3.2. Method

For S independently trained networks (“seeds”), each evaluated with R replicas of the downstream WE comparison, let $x_s = \log(\text{FOM}_{I,s}/\text{FOM}_{E,s})$ denote the log figure-of-merit ratio of the learned coordinate I_θ against the energy-binned baseline E for seed s . The observed across-seed variance,

$$V_{\text{across}} = \frac{1}{S-1} \sum_{s=1}^S (x_s - \bar{x})^2, \quad (2)$$

confounds two sources: genuine between-seed (initialization) variance V_{between} , and within-seed variance V_{within} arising from the finite replica count R used to estimate each x_s . A one-way random-effects decomposition estimates the within-seed component directly, by bootstrap-resampling each seed's own R replicas independently (10^4 resamples per seed in this analysis), and obtains the between-seed component by subtraction, floored at zero:

$$V_{\text{between}} = \max(0, V_{\text{across}} - \overline{V_{\text{within}}}). \quad (3)$$

3.3. Result

Applied to the project's 17-seed sweep of the surrogate-label coordinate at a fixed rarity depth ($R = 10$ replicas per seed): $V_{\text{across}} = 0.2447$ (sd = 0.495), while the mean within-seed variance is $\overline{V_{\text{within}}} = 0.3603$ (sd = 0.600) – *larger* than the raw across-seed variance itself. The subtraction therefore floors exactly at $V_{\text{between}} = 0$ (Table 1). In other words, every seed's own individual measurement is already noisier than the spread between seeds' point estimates; the apparent seed-dependence of the published ratio is consistent with being 100% sampling noise, with no detectable contribution from network initialization.

Quantity	Value
Number of seeds S	17
Replicas per seed R	10
V_{across} (raw, confounded)	0.2447
$\overline{V_{\text{within}}}$ (replica noise, bootstrapped)	0.3603
V_{between} (ANOVA subtraction, floored)	0.0000
Fraction of spread attributable to replica noise	$\geq 100\%$

Table 1: One-way variance decomposition of the log figure-of-merit ratio $\log(\text{FOM}_{I_\theta}/\text{FOM}_E)$ across 17 independently trained networks, $R = 10$ replicas each.

This has a direct practical consequence for compute allocation. A K -network deep ensemble built to average over initialization variance pays $K \times$ the (non-trivial) per-network inference cost – direct profiling of the network used here shows a single forward pass over a WE population of ~ 2000 walkers costs on the order of 200 ms per WE iteration on a single CPU core, dominated by two small convolutional layers – to cancel a variance component this analysis shows is not detectably present. The statistically efficient use of additional compute here is more replicas per existing seed, not more seeds (Table 2), though we note replica cost for a learned coordinate is not free either, since each additional replica still pays the network's per-iteration inference cost.

Strategy	Allocation	Predicted SE
More seeds, fixed $R = 10$	$S = 17$ (current)	0.146
	$S = 50$	0.085
	$S = 150$	0.049
More replicas, fixed $S = 17$	$R = 10$ (current)	0.146
	$R = 80$	0.052
	$R = 320$	0.026

Table 2: Predicted standard error of the pooled mean log-ratio under two compute-allocation strategies, from the fitted ANOVA model of Table 1. The current configuration ($S = 17$, $R = 10$) does not resolve the ratio from unity; both strategies can, at different total compute.

4. Diagnostic 2: seed control and schedule-extension audits

4.1. A silent sign reversal

The 17-seed sweep behind §3 exists because the project's first reported result for this comparison – a single run showing the learned coordinate winning by +14% ($\text{FOM}_{I_\theta} = 4.811 \times 10^{-8}$ vs. $\text{FOM}_E = 4.215 \times 10^{-8}$) – did not

reproduce. A second run of the byte-identical configuration (same cached training labels, same hyperparameters) gave $\text{FOM}_{I_\theta} = 3.484 \times 10^{-8}$, a -17% *loss* against the same FOM_E baseline (Table 3). The two runs’ naive-sampling and energy-baseline figures matched exactly across both runs; only the trained-network-dependent figure differed. The root cause was a missing `torch.manual_seed()` call: the network’s weight initialization was left to the platform’s default (unseeded) random state, so every rerun trained a different network from the same nominal configuration.

Method	Run 1 FOM	Run 2 FOM
naive	1.476×10^{-7}	1.476×10^{-7}
WE[m]	1.173×10^{-7}	1.173×10^{-7}
WE[E]	4.215×10^{-8}	4.215×10^{-8}
WE[I_θ]	4.811×10^{-8} (+14%)	3.484×10^{-8} (−17%)

Table 3: Two runs of an identical configuration, before the seeding fix. naive and WE[E] (energy-binned, no learned network) are bit-identical across runs; only WE[I_θ] (learned network) differs.

This is a narrow, mechanical bug, but its consequence is not narrow: absent the second run, the project would have reported a clean single-comparison win as its headline learned-coordinate result. After fixing the seed and sweeping 17 independent initializations, point-ratio estimates for individual seeds range from -52% to $+112\%$ around unity – consistent with the $+14\%$ and -17% single-run outcomes both being unremarkable draws from the same wide distribution, not evidence of a real effect in either direction from any one run.

4.2. Two schedule extensions independently fail to help

A natural response to a wide, centered-near-unity comparison is to ask whether a finer discretization or a larger sampling budget would let the learned coordinate’s advantage (if any) emerge more clearly. Two such extensions were tested at a deeper rarity target (“beyond=2”), each against the same baselines, and each independently failed:

- **Finer binning** (n_{bins} from 50 to 90, replicas fixed at 10): the learned coordinate’s own relative standard deviation improved only marginally ($0.416 \rightarrow 0.395$) while its cost nearly doubled ($6.90 \times 10^8 \rightarrow 1.19 \times 10^9$), so its own FOM *worsened* ($8.360 \times 10^{-9} \rightarrow 5.363 \times 10^{-9}$). The energy baseline’s cost is structurally insensitive to bin count (it stayed exactly 5.45×10^8 across both configurations), because energy is a discrete, quantized coordinate on this lattice that saturates a small fixed number of occupied bins regardless of n_{bins} – extra bins can only add cost to the continuous learned coordinate, not help it.
- **Larger sampling budget** (WE iterations, `we_iter`, 1000 to 2000, $n_{\text{bins}} = 50$): cost scaled worse than the expected $2\times$ for every method (population grows as the run pushes further into the tail), and every method’s FOM *worsened*, the learned coordinate’s most severely ($\text{FOM}_{I_\theta} : 8.360 \times 10^{-9} \rightarrow 1.698 \times 10^{-9}$, an 80% drop; its own relative standard deviation *increased*, $0.416 \rightarrow 0.540$, despite $3\times$ the sampling).

Table 4 summarizes both. Two structurally different extensions, tested independently, produce the same qualitative outcome: neither rescues the learned coordinate at this depth, and identifying the cost mechanism in each case (a saturating discrete coordinate for binning; a population that grows faster than expected with iteration budget) rules out re-attempting either lever as a productive next step, rather than leaving the question open for a future, more expensive retry.

Extension	Baseline	Extended	Change in FOM_{I_θ}
$n_{\text{bins}}: 50 \rightarrow 90$	8.360×10^{-9}	5.363×10^{-9}	-36%
<code>we_iter</code> : 1000 $\rightarrow$ 2000	8.360×10^{-9}	1.698×10^{-9}	-80%

Table 4: Two schedule extensions at a fixed deeper rarity target (“beyond=2”), both against the same fixed baselines. Both independently worsen the learned coordinate’s own FOM relative to its own prior value.

4.3. Cost-corrected verdict

Reporting a training-objective comparison's cost from simulation sweeps alone, as in Table 3 and the FOM values above, is itself subject to the same blindness raised in §5: the learned coordinate's repeated network inference is uncounted. Recomputing the 17-seed sweep's pooled figure of merit using measured wall-clock time (mean WE[E] native-cost wall time: 397 s over 16 reconfirmed seeds) rather than simulation-sweep cost, with a two-level bootstrap resampling both the seed axis and each seed's replica axis, gives a median ratio of **0.405** (95% CI [0.268, 0.623], excluding unity) – a statistically significant loss for the learned coordinate, in contrast to the simulation-sweep-cost figure, whose 95% CI ([0.84, 1.75] in the project's original analysis) includes unity and would be reported as inconclusive.

5. Diagnostic 3: charge real inference cost, not sweep cost

The two diagnostics above already used wall-clock-corrected costs in their final reported numbers. This section states the general point explicitly and summarizes independent confirmation from the companion paper [13], which uses a different training objective (self-consistency, not surrogate-label) on the same system.

A cost field computed as (number of Monte Carlo sweeps) $\times$ (per-sweep cost) is, by construction, blind to any computation that scales with WE *iterations* rather than sweeps – chiefly, repeated forward passes of a trained network used for binning. Direct instrumented profiling confirms this cost is real and not a batching or threading artifact: the network's forward pass is already correctly batched (one call per WE iteration over the full population, gradient tracking disabled, single-threaded as intended), and its cost is genuine compute-bound CPU time, dominated by two small convolutional layers, that scales with population size.

Measured directly from production run wall-clock timestamps, the learned coordinate's WE stage consistently costs 3–5 $\times$ the energy baseline's wall-clock time across every configuration tested in this project family (range 2.87–4.97 $\times$ across more than twenty independent seed and disorder combinations, both training objectives). In the companion paper's self-consistency-based headline result, charging this real cost converts the comparison from favorable-looking on a count-based reliability metric (which is cost-independent and unaffected) to unfavorable on a cost-normalized efficiency metric: a wall-clock-fair figure-of-merit ratio of 0.33 (95% CI [0.19, 0.58]) [13]. The direction and approximate magnitude of this correction closely matches the surrogate-label objective's independently derived 0.405 $\times$ figure in §4, despite the two training objectives, and the two diagnostics that produced them, being otherwise unrelated – consistent with cost-blindness being a property of the cost accounting, not of either specific learned coordinate.

6. An audit checklist

The three diagnostics above share a structure: each is a specific, inexpensive reanalysis of data already collected, not a request for more expensive simulation, and each is capable of overturning a headline efficiency or reliability claim on its own. We summarize them as a short checklist for reporting learned-coordinate rare-event sampling results.

- (1) **Before ensembling networks to reduce variance, decompose the variance you are trying to reduce.** If a one-way (or richer) variance decomposition shows the between-seed component is small or zero relative to per-seed sampling noise, the ensemble's inference cost buys nothing, and additional replicas of existing networks are the more efficient use of the same compute (§3).
- (2) **Seed every stochastic component of a trained-coordinate pipeline, and confirm determinism by rerunning an identical configuration at least once before reporting a single-run result as a finding.** A cheap rerun is the entire cost of catching a bug capable of reversing a headline sign (§4). Once seeded, test the schedule extensions a reviewer would naturally suggest (finer discretization, larger sampling budget); if they fail, identify the cost mechanism responsible rather than leaving the failure unexplained, since the mechanism determines whether the lever is worth retrying at a different scale.
- (3) **Compute the reported figure of merit's cost term from measured wall-clock time for every method being compared, not from a simulation-sweep count that is blind to learned-component inference.** Where wall-clock measurement is not yet available, treat a sweep-cost-only comparison as provisional and flag it as such, since the correction is not guaranteed to be small (§5).

None of the three checks requires new expensive simulation on the scale of the original comparison being audited: the variance decomposition (§3) reuses already-collected replica data; the seed audit (§4) requires one confirmatory rerun of an existing configuration, cheap relative to a full sweep; and the cost correction (§5) requires only that wall-clock time already be recorded (or, if not, added as a field going forward) rather than any new simulation. We suggest this ordering – variance decomposition, seed and schedule audit, cost correction – as a reasonable default sequence, since each check is comparably cheap and any one of them failing can make the others’ outcome moot for the specific claim at hand.

We do not claim these three are exhaustive, nor that every learned-coordinate rare-event sampling claim in the literature is subject to all three problems; we report only what a systematic reanalysis of one project’s own internal history found. The consistency of direction across two independently developed training objectives on the same underlying system (§5) is, however, some evidence that the cost-blindness problem at least is not an artifact specific to one coordinate design.

7. Conclusions

Using a fully instrumented Weighted Ensemble rare-event sampler for Ising and Edwards–Anderson spin systems, we showed that three inexpensive diagnostic checks – a variance decomposition, a seed and schedule audit, and a wall-clock cost correction – each independently change the reported verdict of a learned-coordinate efficiency or reliability comparison drawn from the same project. A 17-network ensemble’s entire seed-to-seed spread was shown to be finite-replica noise rather than genuine initialization variance. An unseeded initializer silently reversed a headline result between two runs of an identical configuration, and once fixed, two natural schedule extensions were shown, with identified mechanism, to independently fail to rescue the comparison. Charging real wall-clock inference cost rather than simulation-sweep cost reversed an efficiency verdict twice, for two different training objectives on the same system, from an inconclusive or favorable-looking result to a statistically significant loss. None of these findings required new expensive computation; each reused data or compute already paid for by the underlying project. We offer the three checks as a short, concrete checklist that we believe should be applied before, not after, a learned-coordinate rare-event sampling paper’s headline efficiency or reliability claim is finalized.

Data and Code Availability

The scripts implementing the three diagnostics (variance decomposition, seed-sweep and schedule-extension drivers, wall-clock bootstrap correction), along with the underlying run logs and result files referenced in Tables 1–4, are available at <https://github.com/mzaatreh-phs/ising-transport-rare-event>, the same repository as the companion paper [13].

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CRedit authorship contribution statement

M.Y. Alzaatreh: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft. **H. Alsenawe:** Validation, Writing – review & editing.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic) to assist with manuscript organization, language refinement, and software support. After using this tool, the authors reviewed and edited the content as needed,

independently verified the scientific claims and numerical results, and take full responsibility for the content of the publication.

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